

Research paper

Empirical analysis of bifacial photovoltaic modules in high-latitude regions: Performance insights from a field laboratory in Norway

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ABSTRACT

This paper presents a comprehensive empirical analysis of bifacial photovoltaic (bPV) module performance in high-latitude regions, based on data collected from an outdoor field laboratory in Trondheim, Norway. It explores the operational dynamics of bPV systems under environmental challenges such as low irradiation, short daylight hours, low sun angles, and extreme weather conditions, all common in high-latitude regions. The study assesses the performance of various bPV configurations, including vertically oriented modules in east–west (E–W) and south–north (S–N) orientations, fixed south tilted orientations, and orientations equipped with a 2-axis tracking system. Key findings indicate the advantages of mixed orientations in bPV systems, resulting in multiple daily generation peaks that align more closely with consumption profiles. The study demonstrates that bPV modules exhibit a strong response to changes in irradiance than mono-facial modules, particularly to the diffused horizontal irradiance component. This effect is more observed in vertically oriented bPV modules. Yield analysis reveals that spring weather conditions favor PV performance in this region. Additionally, bPV modules show higher performance ratios than monofacial modules in various setups to a different extent, although snow-related reductions are more pronounced in tilted orientations. The results in this work contribute to the optimization of PV systems in regions where low sun angles and snow accumulation are prevalent, emphasizing the potential of vertically oriented bPV modules.

1. Introduction

The global demand for sustainable energy solutions has been steadily increasing, driven by a combination of factors. One of the primary reasons for this surge is the growing awareness of the environmental impacts associated with conventional energy sources. Concerns about climate change have prompted governments, businesses, and consumers to look for clean and sustainable alternatives. Among renewable energy sources, photovoltaic (PV) energy has emerged as a frontrunner, getting significant attention and investment worldwide. According to a report by the International Energy Agency (IEA), the global cumulative installed PV capacity grew to 1.6 TW in 2023 [1]. This growth in PV technology adoption is also becoming notable in high-latitude regions, where conventional PV configurations may result in sub-optimal performance due to the unique climate. In these regions, factors such as low irradiation levels, low sun angles, and significant snowfall during the winter months can lower the performance of conventional PV configurations and PV technologies [2]. In response to the challenges faced by PV systems in high-latitude regions, more research is focusing on adapting and optimizing PV systems for efficient operation in such environments. These works include optimizing PV

installation angles [3], implementing solar trackers [4], using vertical PV configurations [5], and employing bifacial PV modules [6].

In high-latitude regions, PV modules are typically tilted to match the local latitude to ensure maximum absorption of solar irradiation during the winter months. Recent studies have introduced data-driven models to determine the optimal tilt angle, taking into account the effects of snow on energy yield [7]. The consideration of snow factor into optimal angle calculations allows for more accurate annual energy yield estimations and leads to improvements in system performance. In addition to optimizing installation angles, solar trackers have been implemented to enhance PV performance in these regions. Several studies have shown that PV systems using solar trackers generate more energy compared to fixed systems [8]. According to [9] the increase can vary between 15% to 67.65% compared to stationary PV systems. However, this is not always the case. Authors in [10] assessed the performance of a solar tracking PV panel in Montreal, Canada, based on an isotropic sky model. Their study found that solar tracking on cloudy summer days is actually disadvantageous. Moreover, on cloudy winter days, the benefit of solar trackers depends on the presence of ground snow. The study also confirmed that solar tracking could produce up

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to 25% less energy than fixed horizontal modules on cloudy days in spring.

Recently, vertical PV configurations and bifacial PV (bPV) technologies have emerged as promising solutions to optimize energy capture in high-latitude regions. Vertical PV modules can capture low sun angle irradiance and reflected light (albedo) from snow-covered ground better than conventional configurations due to their perpendicular orientation to the sun's rays [11]. On the other hand, bPV systems offer advantages by capturing sunlight from both sides of the panels, including reflected light from snow-covered surfaces during winter months. This capability improves energy yield, utilizing not only direct sunlight but also the reflected component from the snow, thus making bPV systems particularly beneficial for high-latitude regions with snowy winters. Another important advantage of vertical and bPV systems is their ability to produce a generation profile that matches the local consumption patterns better [12]. This is particularly important for vertical bPV modules oriented east–west direction [13]. As efforts to optimize PV systems for high-latitude regions increases, it becomes increasingly important to address the unique challenges these environments present. This requires more research and comprehensive field performance analysis tailored to the specific conditions of high latitudes.

The performance assessment of conventional PV systems has long been a subject of interest. In recent years, bPV systems have received increased attention. In all the studies reviewed in the scope of this paper, bPV systems outperformed monofacial systems across various scenarios and climatic zones. However, the reported performance gains vary in each case. Authors in [14] reported that the year-round performance of bPV systems in cities across India's five climatic zones is 30.54%–34.93% higher than that of monofacial PV modules. Similarly, a study by [15] highlighted a maximum energy gain of 16.3% for bPV systems in desert climates. Meanwhile, [16] found that bPV systems can generate up to 12% more energy at latitudes below 65° with this advantage increasing to 71% at latitudes above 65°. Additionally, simulation results indicate that bPV modules installed on rooftops in Australia can achieve energy yield gains of up to 22.6% [17]. Nevertheless, the majority of research efforts understandably concentrated on systems situated in hot and moderate climate conditions, owing to the well-established maturity and widespread adoption of the technology in these regions. It is important to acknowledge that the insights gained from these studies may not be directly applicable to PV systems operating in high latitude and cold regions. Therefore, the need for research specific to high-latitude regions is both timely and important. Such research will help address existing knowledge gaps and offer unique insights into the performance of bPV in these regions.

Given the rapid expansion of PV installations in high latitude regions in recent years, there has been a rise in interest to explore into and analyze the performance of these PV systems. In their study, researchers in [12] demonstrated that bPV systems positioned vertically in high latitude regions possess a distinct advantage, as they capture solar irradiation from lower sun angles over an extended duration. Moreover, their findings indicate that vertically oriented east–west bPV systems exhibit promising potential, closely aligning energy production with consumption profiles. A similar work has also been carried out by [18], focusing on devising a framework to identify a configuration that optimizes PV energy for family homes in Canada. Their results indicate that a south-west facing PV system, inclined at a tilt angle of 10° above or below the local latitude, enhances self-consumption compared to conventional installation practices in high latitude regions.

The work by [7] introduces a data-driven prediction model designed specifically for high-latitude regions, aiming to identify optimal PV configurations. Their findings underscore the significance of accounting for soiling caused by snow in assessing PV performance, revealing an average annual yield loss of 14.7%. The effectiveness of vertically mounted bPV systems in the context of low-rise residential settings was analyzed by [11]. Their findings reveal that vertical bPV installations positioned along the ridges of unshaded roofs, specifically aligned with

the north–south axis, outperform monofacial PVs (mPVs) on east- and west-facing roofs by 20%–30%. Upon thorough examination of the aforementioned literature, it is evident that much uncertainty still exists regarding the performance of bPV systems in high latitude and cold regions. This underscores the necessity for further research in this domain.

The literature identifies two key limitations in assessing bPV operation in high-latitude regions. First, there is a shortage of proven field data specific to these regions, which limits the accurate evaluation and assessment of these systems. Second, analysis based on existing simulation tools for bPV systems lack sufficient accuracy to capture the unique challenges of high latitudes. This limitation can undermine the applicability of such approaches. This work aims to address the former issue, addressing the gap in available empirical evidence. Building upon prior research, it provides a comprehensive understanding and insights into the performance of bPV systems in high latitude and cold regions. It uses experimental data collected over a one-year period from a field laboratory in Norway and evaluate the system using various yield and performance indicating metrics. This work is comprehensive in that it evaluates both mPV and bPV modules and considering various feasible orientations and configurations, including tilted, vertical, and 2-axis tracking systems. Furthermore, it uses high-resolution field measurements to capture the dynamics of both PV power and weather parameters. The findings presented in this paper offer new insights and contribute to the limited body of knowledge concerning the performance of bifacial PV systems in regions characterized by high latitudes and cold climates.

The subsequent sections of the paper follow this structure: Section 2 details the experimental system, providing information on the case study area, including the types and orientations of the PV modules, the weather measurement system, and the implemented data collection. In Section 3, various assessment methods are discussed for quantifying the performance of different PV technologies and configurations. This section also discusses the uncertainty related to this experimental work. Section 4 presents the main findings and discussions derived from the study, offering a comprehensive analysis of key observations and insights. Finally, Section 5 outlines the main conclusions drawn from this work, providing a concise summary and suggesting potential areas for future research.

2. Experimental system description

The experimental investigation in this paper is carried out at Alpha Centauri, a field laboratory situated at Gløshaugen in Trondheim, Norway (63.41 N, 10.40 E) [19]. Alpha Centauri is an outdoor test facility specifically designed for the rigorous testing and characterization of PV modules in the challenging Nordic climate. The Alpha Centauri lab include the following important components:

- Eight bifacial PV modules mounted on four racks equipped with two-axis control.
- Individual data loggers to measure important parameters such as short-circuit current, open circuit voltage, output power, module temperature, and other relevant variables.
- A weather station, capturing a range of environmental parameters.
- A specialized module characterization device.

A brief discussion of the important components is included below.

2.1. Photovoltaic modules

The PV metering system under consideration in this paper consists of eight bifacial PV modules, organized into four racks, each containing two modules (refer to Fig. 1). The racks are positioned 6 meters apart horizontally and are elevated 1 meter above the ground. Each rack represents a unique configuration, yielding four different orientations



Fig. 1. Alpha Centauri: Field laboratory for testing solar PV modules in Trondheim Norway. With meteo station and irradiation measurements.

for analytical purposes. For this experimental campaign, the first rack (MS01/02) is configured vertically in the South–North (S–N) direction, while the second rack (MS03/04) is fixed in a south-facing position with a tilt angle of 44° . The third rack (MS05/06) operates in tracking mode, and the fourth rack (MS07/08) is set up vertically in the East–West (E–W) direction. Within each rack, one of the bifacial modules is equipped with a rear-side cover, allowing it to perform essentially as a monofacial PV module. However, this assumption should be approached with caution, as covering the rear side may alter thermal and electrical characteristics, potentially leading to inaccuracies in performance evaluations. Nonetheless, for the area considered in this paper, characterized by low temperatures and minimal thermal stress, the potential negative effects of altered thermal properties due to rear-side covering are minimal, making the assumption more valid.

The configurations in the experimental setup allow a comprehensive assessment of diverse module orientations and both mono and bifacial module types. Each module consists 60 mono-crystalline silicon PERC (Passivated Emitter and Rear Contact) solar cells and the specific electrical details for each bifacial PV module are provided in Table 1. These values are from a flash test on each module at Standard Test Conditions (STC).¹ The modules are denoted as MS0X, where $X = 1, 2 \dots 8$. Even-numbered modules feature a backside cover essentially operating as monofacial modules.

2.2. Weather station

The weather station at Alpha Centauri is equipped with two pyranometers (Kipp & Zonen CMP10). The first pyranometer captures Global Horizontal Irradiance (GHI), providing data on the total solar

radiation received from the entire hemisphere above the horizontal plane. The second pyranometer measures Diffuse Horizontal Irradiance (DHI), the scattered solar radiation component. Additionally, the station has a pyrheliometer (Kipp & Zonen CHP1) that measures the Direct Normal Irradiance (DNI), the solar radiation that arrives on a surface perpendicular to the sun's direction. The measurement from these instruments allow understanding of the solar radiation dynamics. In addition to the solar radiation measurements, the station has instruments (Thies CLIMA) for measuring ambient temperature, wind speed and direction, relative humidity, and precipitation. These sensors enable the collection of a broad range of meteorological data that are important for PV applications.

2.3. Data collection

Each PV module is equipped with individual data loggers with a time resolution of 10 s (ISET[®]-mpp meter). These loggers capture essential parameters, including short-circuit current, open-circuit voltage, output power, and module temperature. Furthermore, the bPV modules are equipped with high-quality solar radiation ISET sensors. These sensors measure plane-of-array radiation from both the front and back sides of the modules. In this study, a 12-month data collection process was conducted, spanning from July 2022 to June 2023. This time frame includes a diverse range of weather conditions and the required seasonal variations. The data collection is done through a PV monitoring and acquisition platform (SOL.Connect[®] PV Analyzer). The time series data, initially collected at 10-s intervals, is subsequently down-sampled to 15-minute intervals. This down-sampling allows a practical and detailed examination of temporal patterns and fluctuations. A concise summary of the gathered meteo data is presented in Table 2, accompanied by visual representations in Figs. 2, 3, and 4. Solar radiation is high during the summer months and low in the winter. The annual average hourly temperature is 6.1°C , with the

¹ This corresponds to irradiance level of 1000 W/m^2 , temperature of 25°C , and 1.5 air mass.

Table 1

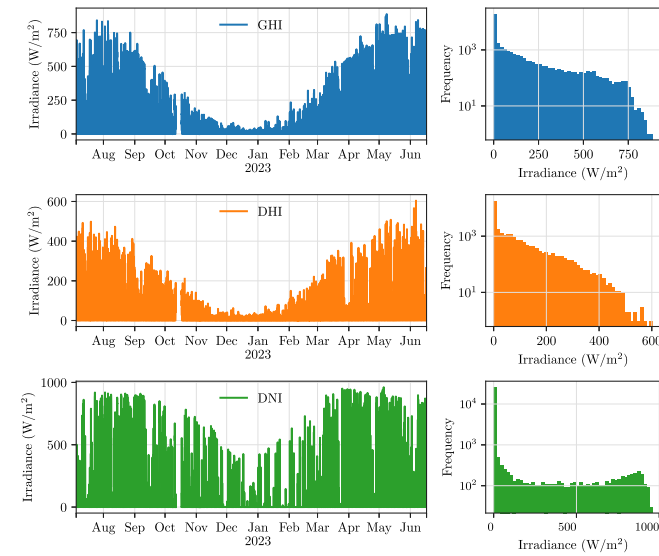
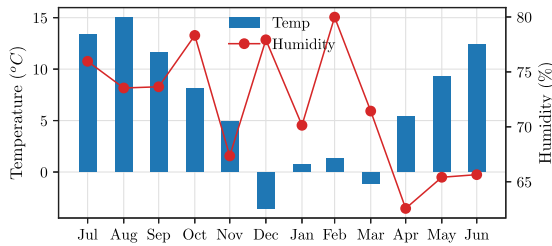
Electrical specifications and configuration of PV modules at Alpha Centauri field laboratory. These specification are based on a flash test.

Module	I_{sc} (A)	V_{oc} (V)	I_{mp} (A)	V_{mp} (V)	P_{mp} (W)	FF (%)	Tilt (°)	Orientation	Back cover
MS01	9.25	40.4	8.8	33.4	294.5	78.9	90	S-N	×
MS02	9.24	40.4	8.8	33.5	294.8	79.0	90	S-N	✓
MS03	9.28	40.4	8.8	33.5	296.0	78.9	44	Fixed S	×
MS04	9.25	40.4	8.8	33.5	295.1	78.9	44	Fixed S	✓
MS05	9.25	40.4	8.8	33.5	295.0	78.8	^a	Tracking	×
MS06	9.29	40.4	8.8	33.5	295.3	78.7	^a	Tracking	✓
MS07	9.28	40.4	8.8	33.5	295.5	78.9	90	E-W	×
MS08	9.26	40.3	8.8	33.5	294.8	78.9	90	E-W	✓

^a Equipped with 2-axis tracking.**Table 2**

Statistical summary of the collected meteo data for the study period.

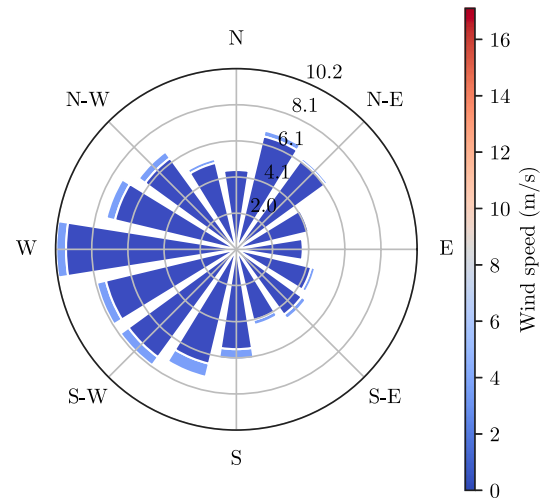
	GHI (W/m ²)	DNI (W/m ²)	DHI (W/m ²)	Humidity (%)	Temperature (°C)	Wind (m/s)	Precipitation (mm)
Mean	88.48	95.89	49.07	72	6.10	1.05	0.8
Min	0.00	0.00	0.00	23	−15.80	0.00	0.00
Max	887.18	961.65	604.25	101	26.20	17	32.70

**Fig. 2.** Components of monthly average solar irradiance and frequency distribution: GHI, DHI, and DNI.**Fig. 3.** Monthly average temperature and humidity for the study period.

lowest temperatures recorded in December. Wind speeds can reach up to 17 m/s, but the area is primarily characterized by relatively low-speed winds, especially those coming from the west.

3. Assessment method

The following set of assessment metrics and parameters are used to evaluate and quantify the performance of various PV modules and configurations. The assessment metrics are grouped into three main factors: correlation study, those pertaining to electrical characteristics

**Fig. 4.** The distribution of wind speed and direction for the study period.

and those related to bifacial gain. This grouping allows for a thorough examination of the performance of bPV system under consideration.

3.1. Correlation study

Understanding the relationship between weather parameters and the output power from PV modules is a critical aspect of PV energy system analysis. In this study, the Pearson Correlation Coefficient (PCC) is used, a well-established and widely accepted statistical metric. It is used for studying the correlation between various meteorological factors, including GHI, DNI, DHI, temperature, and other parameters, in relation to the output power from each PV configuration. The PCC between two variables x and y is calculated using

$$PCC_{x,y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

where n is the sample size, x_i , y_i are the individual data points and $\bar{x}$, $\bar{y}$ are the mean values of sample.

3.2. Performance and electrical characteristics

Specific yield, performance ratio, fill factor, and IV-curves are used as a performance and electrical characteristics indicating metrics for different PV configurations in this work.

3.2.1. Specific yield

Specific yield refers to the energy output of a PV system per unit of installed capacity. It is expressed as the ratio of kilowatt-hours (kWh) of electricity generated to kilowatt-peak (kWp) of installed PV capacity over a specific period. Factors such as geographical location, weather conditions, PV technology and configuration, and shading can affect the yield from a PV module.

3.2.2. Performance ratio

The performance ratio (PR) indicates the overall efficiency of a PV module, accounting for losses attributable to factors such as temperature, mismatch, shading, angular effects, and degradation. It is calculated as the ratio of the actual energy output to the theoretically possible energy output under STC conditions, i.e.,

$$PR = \frac{\left(\frac{P_{mpp}}{P_{mpp,STC}} \right)}{\left(\frac{I_{PoA}}{I_{PoA,STC}} \right)} \times 100\% \quad (2)$$

where P_{mpp} is the maximum power at real or ideal (STC) conditions and I_{PoA} is the plane-of-array irradiance of the module at real or STC conditions (i.e., 1000 W/m²).

3.2.3. Fill factor

Fill factor (FF) is a measure of how effectively a PV module converts solar energy into electrical power. It is the ratio of the actual maximum power to the product of the open circuit voltage (V_{oc}) and short circuit current (I_{sc}). It is calculated as

$$FF = \frac{P_{mpp}}{V_{oc} \times I_{sc}} \times 100\% \quad (3)$$

where P_{mpp} is the output power at MPP.

3.2.4. Characteristics curves

The current voltage (IV) and power voltage (P–V) curves are important for understanding the electrical characteristics and performance of a PV module. They are affected by factors such as solar irradiance, ambient/cell temperature, and wind speed. These curves can reveal issues such as shading, dirt, or malfunctioning cells. Changes in the shape of the expected curve or reduced performance can indicate potential problems in the PV module.

3.3. Bifacial gain

Bifacial PV modules use both the front and rear surfaces of the module to enhance overall energy capture. This can be quantified using bifacial coefficients that can be defined on optical, module, and system level [20]. At the optical level, bifacial coefficient is calculated based on the irradiance received by both sides of the module. On the module level, it accounts for the difference in efficiencies between the front and rear side module. Furthermore, on the system level, the bifacial coefficient which is widely known as the bifacial gain is quantified by comparing the overall performance of the bifacial PV module with a monofacial PV module having the same orientation and technology. It shows the additional power derived from sunlight striking the rear side of the PV modules.

3.3.1. Optical bifacial coefficient

The optical bifacial coefficient (BC_o) is an important metric that shows the average irradiance gain achieved through the rear surface of a bPV module. This coefficient quantifies the theoretical maximum potential of the rear side of the module for solar energy capture. It indicates how efficiently the rear side of the bifacial module can contribute to the overall energy yield. It is calculated as

$$BCo = \frac{I_{rear}}{I_{front}} \times 100\% \quad (4)$$

where I_{rear} and I_{front} are the rear and front side measured irradiance in W/m².

Table 3

Main measurement instruments and the associated uncertainties considered in this paper.

Parameter	Equipment	Uncertainty (%)
Irradiance	Kipp & Zonen CMP10 Pyranometer	3.21
Irradiance	Kipp & Zonen CHP1 Pyrheliometer	1.24
Current/Voltage	SOL.Connect ISET®- mpp meter	1.41
Logger	SOL.Connect	0.2

3.3.2. Module bifacial coefficient

The module bifacial coefficient (BC_m) indicates the increased average irradiance obtained from the rear surface of a bifacial module. Unlike optical bifacial coefficient, this coefficient takes into account the efficiency of the rear side, which is typically lower than that of the front side. This particular factor is denoted as the bifaciality factor (ϕ) and is specified in the module's datasheet ($\phi = 0.65$). The module bifacial coefficient is calculated as

$$BCm = \frac{\phi \times I_{rear}}{I_{front}} \times 100\% = \phi \times BCo \quad (5)$$

3.3.3. System bifacial coefficient/gain

System bifacial gain (BG_s) refers to the ratio between the overall net energy output of a bPV module and that of a mPV counterpart, both operating under identical conditions and configurations. This metric is influenced by several factors, including the albedo of the ground surface, the tilt and height of the PV modules, and the environmental conditions in the surrounding area. It is calculated as

$$BGs = \frac{E_{bPV} - E_{mPV}}{E_{mPV}} \times 100\% \quad (6)$$

where E_{bPV} and E_{mPV} are the net energy output from a bifacial and monofacial PV module respectively.

3.4. Uncertainty analysis

Uncertainty analysis quantifies the level of confidence in experimental data by identifying and combining potential sources of error, defining a range within which the true value is likely to fall. It is important in experimental studies like this one, as it ensures the reliability of the analysis and enables conclusions based on measurements with quantified accuracy. The primary sources of uncertainty considered in this paper include uncertainties associated with the measurement of irradiance, current, and voltage, as well as those related to the data logging unit. These uncertainties are summarized in Table 3.

It is important to note that the uncertainty values for the pyranometer and pyrheliometer shown in Table 3 represent total uncertainty, which includes contributions from non-stability, non-linearity, spectral selectivity, and temperature response. Assuming a uniform (or rectangular) distribution for the uncertainty, a 95% confidence level, and a coverage factor of (k), the combined uncertainty in the analysis of yield and power in this work can be calculated as

$$U = k \times \sqrt{\left(\frac{U_{pyra}}{\sqrt{3}} \right)^2 + \left(\frac{U_{pyrh}}{\sqrt{3}} \right)^2 + \left(\frac{U_{IV}}{\sqrt{3}} \right)^2 + \left(\frac{U_{logg}}{\sqrt{3}} \right)^2} \quad (7)$$

where U_{pyra} , U_{pyrh} , U_{pyra} , and U_{pyra} are uncertainties from the pyranometer, pyrheliometer, current–voltage measurement, and the data logger, respectively. For a coverage factor of 2, the above equation results a combined uncertainty of 4.3%. The uncertainty analysis approach used in this paper is based on the work of [21,22]. A limitation of the above analysis is that it does not account for uncertainties related to environmental conditions, such as soiling from dirt and, more importantly, snow accumulation during winter months, which can impact

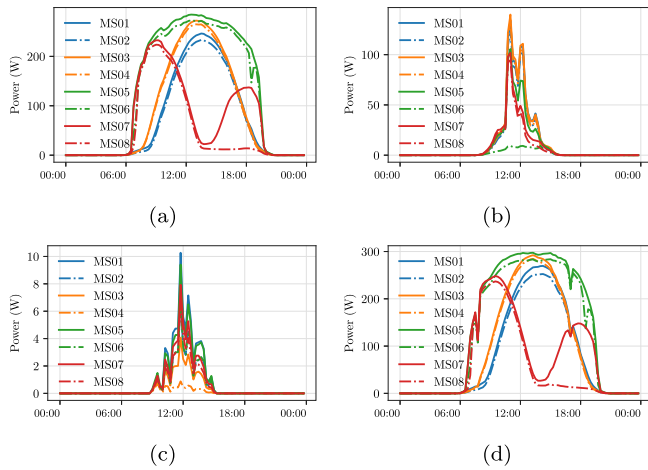


Fig. 5. Typical generation profile of Alpha Centauri PV system for clear sky and cloudy days. (a) Sep-06-2022. (b) Nov-06-2022. (c) Dec-06-2022. (d) Apr-06-2023.

irradiance measurements and, consequently, yield estimates. Future assessments should consider the effects of soiling and seasonal snow accumulation on irradiance measurements to provide a more practical and accurate representation of uncertainty.

4. Results and discussion

This section summarizes and discusses the main findings of this empirical work based on the assessment methods discussed in the previous section. Data spanning over a period of 1 year is used for this study. This section starts with a general discussion of the output power profile from the system. The power output profile of a PV module is dependent on the physical orientation of the system, including both tilt and azimuth angles, as well as the type of module. This is illustrated in Fig. 5, which presents data from 4 randomly selected days across 4 distinct seasons. Upon closer inspection of the plots, it is evident that there are multiple generation peaks throughout the day. Such a multiple generation peak profile is highly desirable as it aligns more closely with consumption patterns, effectively shifting the PV power profile in time to better match the daily consumption patterns. This aspect is particularly important in regions where building integrated photovoltaic (BIPV) systems, which are ideal for such mixed orientations, hold significant potential for PV expansion. Norway serves as a prime example of such regions, where optimizing land use is critical for better PV adoption.

Fig. 5 also illustrates that the output power is significantly reduced during winter days for all configurations and PV types. This observation aligns with expectations, given the region's characteristics of significant snowfall, low irradiance, and short sunlight hours during winter. It is also interesting to note that PV modules with 2-axis sun tracking produce more and bPV modules have better yields than their monofacial counterparts. This latter aspect is particularly notable in vertical E-W configurations, where bPV modules exhibit two generation peaks – one in the morning and one in the late afternoon – unlike mPV systems, which have only one peak in the morning.

4.1. Correlation analysis

The correlation between GHI, DHI, DNI, and measured PV power for both mono-facial and bifacial modules, as well as various configurations, is important for understanding system performance under diverse irradiance conditions. The results of this analysis are summarized in Figs. 6(a) and 6(b) for the mono-facial and bifacial cases, respectively. The figures reveal that GHI and DNI exhibit a stronger correlation

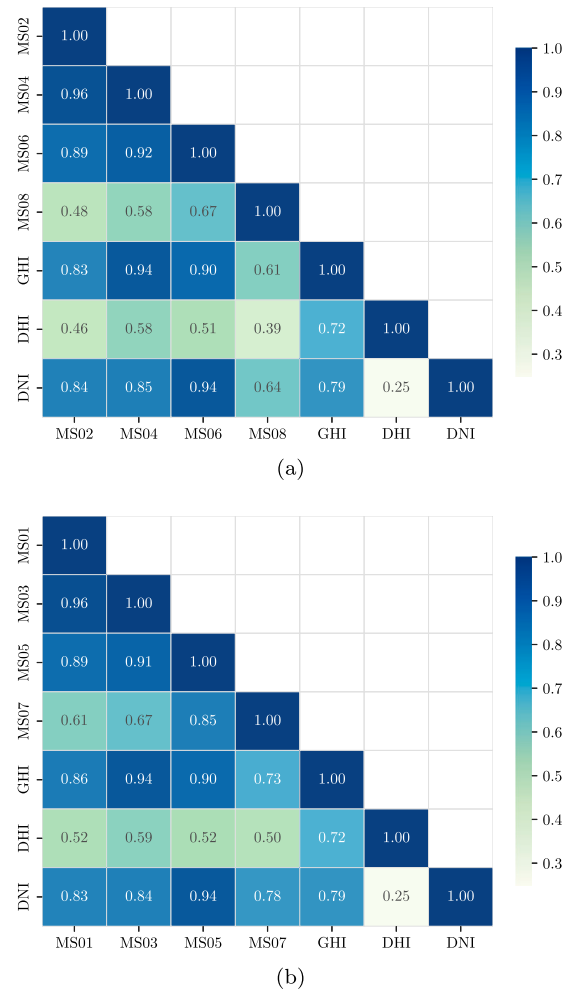


Fig. 6. Correlation analysis between the solar irradiation components and the corresponding measured power for monofacial and bifacial modules in different configuration (a) Mono-facial PV modules. (b) Bifacial PV modules.

with output power, regardless of the PV module type or configuration. Conversely, a contrasting trend is observed for DHI. Notably, bifacial PV modules demonstrate a slightly stronger correlation across all configurations. The strong correlations between GHI, DNI, and PV output for both systems are primarily due to the significant impact of the direct solar irradiation component in PV power generation. Both irradiance measurements (GHI and DNI) include the direct beam, which PV modules convert most efficiently into electricity. On the other hand, the observed correlation with DHI for the two systems, arises from the scattered nature of this irradiance, which is less efficiently captured by monofacial PV modules.

The following important observation can be obtained from these results:

- GHI and DNI exert similar influence on the performance the PV systems, irrespective of the configuration or technology. This is evident from the high and fairly similar correlation coefficients both for mono facial and bifacial modules.
- The results suggest that bifacial PV modules are more sensitive to changes in irradiance, making them susceptible to variations common in the study area. This is evident from the slightly higher correlation coefficients for the bifacial modules for all solar irradiance components most of the time.
- The results also suggest that DHI plays an important role in optimizing bifacial PV module performance in high-latitude regions.

Table 4

Yield across the season in various PV configuration in kWh. The values in bold indicate total values.

Month	Vertical SN		Tilted		Tracking		Vertical EW	
	MS01	MS02	MS03	MS04	MS05	MS06	MS07	MS08
09-2022	27.44	24.77	31.50	30.34	43.45	40.47	23.03	15.13
10-2022	13.17	12.10	12.76	12.90	15.68	15.34	8.16	5.65
11-2022	11.91	11.19	9.28	9.35	12.11	11.94	3.87	3.09
Autumn	52.53	48.05	53.54	52.59	71.24	67.74	35.07	23.88
12-2022	4.94	4.56	1.80	1.55	4.85	4.36	1.17	0.88
01-2023	2.67	2.26	2.19	1.96	2.69	2.40	1.33	0.99
02-2023	9.10	7.79	7.98	7.35	10.02	8.97	5.31	3.73
Winter	16.72	14.60	11.96	10.87	17.56	15.73	7.82	5.59
03-2023	29.90	25.43	29.26	26.61	41.38	36.45	23.81	16.54
04-2023	34.64	30.09	41.20	39.35	57.21	52.43	33.71	22.36
05-2023	32.94	26.96	44.54	42.20	61.39	55.31	36.78	23.31
Spring	97.48	82.49	114.99	108.16	159.98	144.20	94.29	62.22
06-2023	17.63	13.74	24.59	23.13	34.03	32.80	21.22	12.93
07-2022	18.72	14.18	25.56	24.07	31.83	29.45	19.88	11.74
08-2022	30.37	25.82	37.32	35.49	48.40	44.28	32.57	23.11
Summer	66.71	53.75	87.47	82.69	114.26	106.53	73.67	47.78
Total	233.44	198.89	267.97	254.31	363.04	334.20	210.85	139.46

This is particularly true for vertically oriented systems *MS01* and *MS07*. The bifacial modules *MS01* and *MS07* have higher correlation coefficients of 0.52 and 0.52 for DHI compared to *MS02* and *MS08* which has 0.46 and 0.39 respectively.

It is important to acknowledge that the direct application of the PCC in evaluating the relationships between PV output and environmental factors may lead to potentially misleading interpretations. Specifically, the PCC assumes a linear relationship between variables, which may not adequately capture the true nature of the associations between PV power and environmental factors. Additionally, environmental factors often exhibit multicollinearity, which complicates the assessment of their individual effects on PV output. Therefore, while the above analysis based on PCC highlights very important correlations, the results should be interpreted with caution.

4.2. Yield assessment

This section investigates the yield performance of the PV system in Alpha Centauri. Through this analysis, the impacts of varying system designs, PV types, and seasonal variations on yield outcomes are discussed. The results of this analysis are summarized in Table 4. This table shows the yield of the different PV configurations in Alpha Centauri across various seasons for a one-year period. In general, the highest yield (363.04 kWh) is obtained from a bPV module equipped with a 2-axis tracking system (*MS05*). This system produces 9% more yield compared to the second-best configuration, a 2-axis tracking mPV module (*MS06*). On the other hand, a mPV module with an E–W configuration (*MS08*) has the lowest yield, 139.46 kWh. The observed performance of the bifacial PV module with a 2-axis tracking system (*MS05*) is due to its ability to capture irradiance from both direct and reflected light while continuously optimizing its orientation to the sun. In contrast, the vertical monofacial module with a fixed E–W configuration (*MS08*) has inherent limitations, capturing solar irradiation primarily during the morning hours, which leads to lower yields. A consistent trend emerges when considering seasonal variations. Across all PV configurations, there is a significant decrease in generation during the winter season. Spring is the peak season for yield across all configurations, followed by summer, and autumn. This pattern suggests that springtime conditions offer the most optimal environment for PV system in high latitude regions. Despite fluctuations throughout the year, the order of seasonal performance remains consistent across different PV setups. This indicates the importance of taking seasonality into consideration while optimizing PV systems in this region.

The analysis of yield potential, comparing bifacial and monofacial PV modules across various configurations shows interesting results.

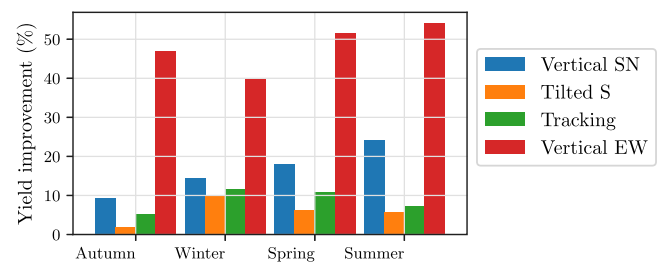


Fig. 7. Seasonal yield improvement potential due to bPV modules compared with mPV counterparts for different configurations.

This analysis is shown in Fig. 7. Firstly, the E–W bifacial configuration exhibits the highest potential throughout the year, with the peak being in summer at 54%. This is because, during the summer, the sun's higher and longer path provides more sunlight hours, allowing a vertical E–W bifacial configuration to capture solar irradiation during the extended mornings and evenings. Secondly, in winter, both tilted-fixed and tracking configurations show significant gains from bifacial modules, with increases of 10% and 12% respectively. This is directly related to the increased ground reflectivity due to snow cover (albedo) and the higher proportion of diffused solar irradiation during winter, both of which improve rear-side irradiance capture. However, what stands out is the comparatively lower improvement in the peak season for tracking and fixed-tilted configurations compared to vertical setups. This is possibly because during peak seasons like spring, tracking and fixed-tilted systems are already operating near their maximum potential by directly facing the sun and capturing most of the direct irradiance, so the incremental benefit from rear-side capture is proportionally smaller. Additionally, the ground albedo is very low during this period. These observations underscore the importance of optimizing bPV module and configuration choices across seasons in high latitude regions.

In order to provide comprehensive performance evaluation and enable comparative analysis across literature, the results from specific yield assessments are shown in Figs. 8(a), 8(b), 8(c), and 8(d). These results consider the system's capacity, providing a normalized measure of performance. This enables a better understanding of how different PV configurations in various locations and using different technologies impact yield. It can be seen from these figures that the specific yield has a similar pattern to the yield of the system discussed above. This is because all the PV modules considered in this work have the same technology and capacity. The E–W configured mPV module (*MS08*) has the lowest specific yield in all seasons of the year: 2.97, 56.06,

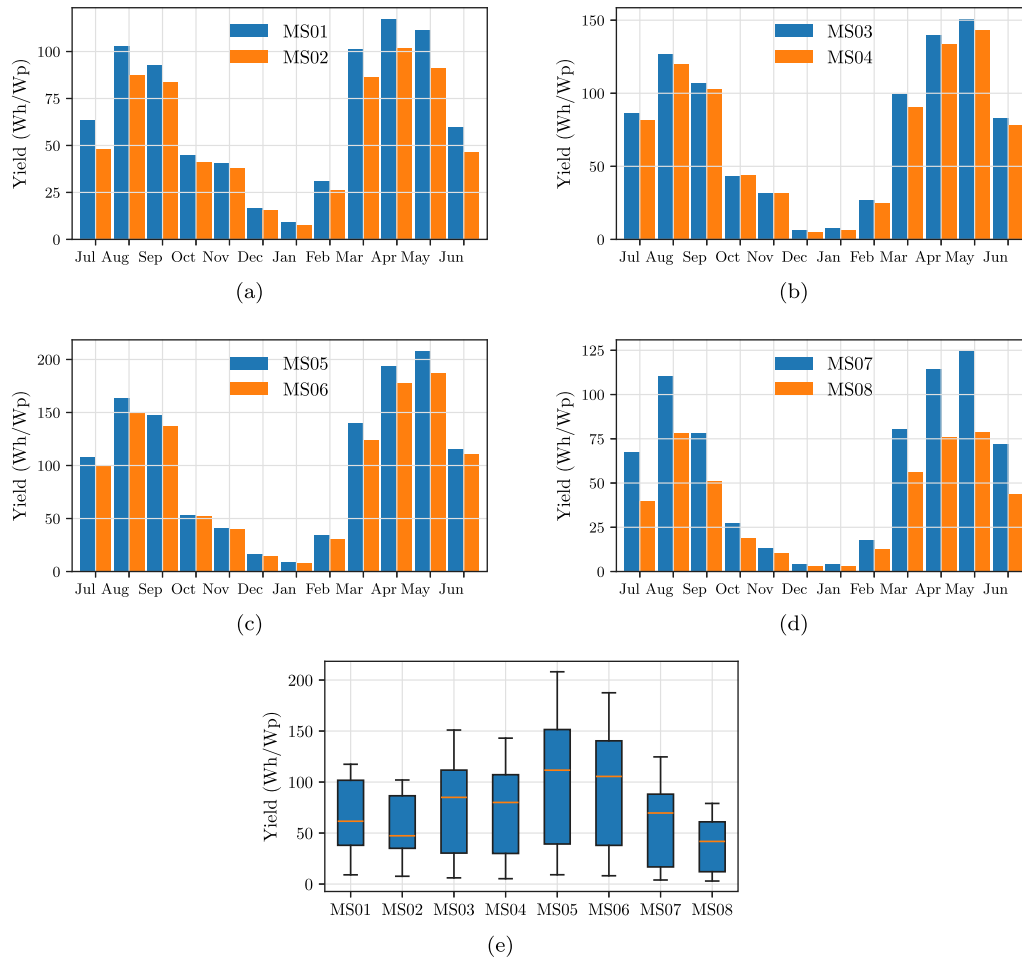


Fig. 8. Specific yield assessment of monofacial and bifacial PV modules for different configurations. (a) Vertical S-N. (b) Tilted S. (c) Tracking. (d) Vertical E-W. (e) Annual spread of the specific yield using box and whisker plot. The box represents the middle 50% of the data. The whiskers show the smallest and largest values and the orange line marks the median.

39.79, and 10.46 Wh/Wp in winter, spring, summer, and autumn, respectively. Conversely, the bPV module with 2-axis tracking has the highest specific yield in all seasons.

A concise visual summary of the distribution of the specific yield in all months of the year for different configurations is provided using a box-and-whisker plot shown in Fig. 8(e). It can be seen that the vertical configurations have relatively narrower box lengths and shorter whiskers. These suggest that less variability and uniform behavior in terms of performance is observed in these configurations across different seasons. A possible explanation for this observation is that vertically oriented PV modules are less sensitive to seasonal variations in high latitude regions. This observation suggests that vertical PV systems can provide more predictable and stable energy output throughout the year. Conversely, an opposite behavior can be observed for fixed tilted and tracking PV modules.

4.3. Performance ratio

The performance ratio measures the overall efficiency of a PV module by taking into account various system losses. Table 5 presents the results of this analysis. In the calculation of PR for bifacial modules, bifaciality factor ($\phi = 0.65$) and albedo of the roof (0.13) was used to correct the Pmpp at STC. Generally, the bifacial modules have a higher performance ratio than monofacial modules. The higher performance ratio of bifacial modules is primarily due to their ability to capture additional solar irradiance from the rear side, leading to higher overall energy generation. However, an exception to this trend is observed in

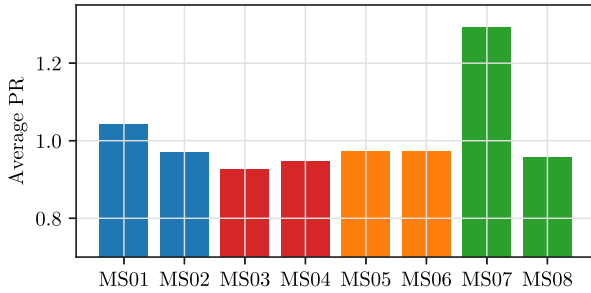
the tilted south-facing configuration (MS03/04). In this configuration, the monofacial module (MS04) has a slightly higher performance ratio than its bifacial counterpart (MS03). This is possibly due to the tilt angle of the configuration (44°). The tilt angle in this case is optimized to maximize the capture of direct sunlight on the front side of the panel. This optimization may have significantly negatively affected the energy capture from the rear side for the bifacial module.

Another important finding is that vertically oriented bifacial modules exhibit high PR during the summer, while fixed south and tracking modules perform better in the colder seasons (February and December for fixed south and tracking configurations, respectively). This is consistent with the observation made in the yield analysis in the previous subsection, and the same underlying reasoning applies here. The east-west oriented bifacial PV module (MS07) has the highest PR of 1.42 in July. The yearly average PR of all the PV modules was evaluated based on the results in Table 5. This analysis is illustrated in Fig. 9. From this figure, it can be seen that the east-west configuration has the highest yearly average PR (1.29), while the tilted south bifacial module has the lowest average value (0.92). This figure also shows the potential benefits of using bifacial modules instead of monofacial modules for different configurations. Based on this figure, it can be concluded that no significant benefit in terms of PR is achieved by opting for bifacial modules in tilted south and sun tracking mode configurations, unlike the north-south and east-west vertical configurations. A possible explanation for this is that tilted south and sun-tracking configurations are optimized to maximize direct annual solar irradiation on the front side of the module. In such setups, the backside receives minimal

Table 5

Monthly summary of the performance ratio for monofacial and bifacial modules with different configurations.

Month	MS01	MS02	MS03	MS04	MS05	MS06	MS07	MS08
2022-07	1.122	0.922	0.940	0.960	0.945	0.948	1.422	0.911
2022-08	0.992	0.915	0.911	0.939	0.932	0.924	1.184	0.911
2022-09	0.953	0.933	0.909	0.950	0.932	0.941	1.306	0.931
2022-10	0.963	0.960	0.890	0.976	0.927	0.983	1.294	0.972
2022-11	1.001	1.017	0.923	1.009	0.966	1.032	1.131	0.978
2022-12	1.103	1.103	0.657	0.617	1.105	1.077	1.117	0.904
2023-01	1.107	1.015	1.026	0.995	1.016	0.983	1.283	1.030
2023-02	1.103	1.023	1.046	1.046	1.057	1.026	1.343	1.022
2023-03	1.056	0.974	1.014	1.000	1.031	0.985	1.398	1.053
2023-04	1.000	0.942	0.933	0.967	0.955	0.949	1.316	0.947
2023-05	1.048	0.930	0.937	0.962	0.948	0.926	1.331	0.915
2023-06	1.065	0.900	0.919	0.938	0.861	0.900	1.372	0.906

**Fig. 9.** Yearly average performance ratio for monofacial and bifacial PV modules for various configurations.

irradiance, so the additional energy gain from the rear side is small, resulting in no significant improvement in the PR.

4.4. Characteristics curves

The IV–PV characterization of the PV modules at Alpha Centauri was conducted under varying irradiance and temperature conditions at different times of the day. The resulting IV curves and corresponding values are shown in Fig. 10 and Table 6 respectively. The table and figure show that the bPV and mPV modules with the same configuration have fairly similar characteristics. However, notable differences are there across the different configurations. The observed similarities and differences are fundamentally due to how installation configurations influence the additional energy gain potential of the rear side of bifacial modules.

The impact of solar irradiation change on the operating point of the modules and on module's temperature distribution was analyzed for the various configurations. When the DNI changes from 677 at 9:00 to 828 W/m² at 12:00, the largest increase in solar irradiation received by a module was observed in the bifacial module with a tilted south orientation (MS03). The received solar irradiation increased by 536 W/m², with a corresponding module temperature rise of 11.7 degrees. This resulted in an increase in the I_{mp} by 5 A and a decrease in the V_{mp} by 1.65 V, leading to a total P_{mp} increase of 147 W. This can be observed by the change of the characteristic curve shown in Fig. 10 from its position at 9:00. In contrast, the E–W oriented modules showed an opposite pattern. During the same period, the irradiation received by these modules decreased by 413 and 388 W/m², and the module temperature dropped by 9.8 and 9.5 degrees for MS07 and MS08, respectively. This resulted in a decrease in the P_{mp} by 110 and 104 W, respectively.

Similarly, when the DNI changes from the value at 12:00 to 837 W/m² at 14:00 after solar noon, the most significant change in received irradiation was observed in modules configured vertically in an East–West (E–W) direction. Specifically, there was a decrease of 340 W/m² and 323 W/m² for monofacial and bifacial modules, respectively. This

decrease resulted in a corresponding module temperature change of –5.2 degrees and –1.8 degrees. Consequently, a net decrease of output power by 91.17 W for the MS07 and 92.98 W for MS08 was observed. This effect is clearly illustrated in the IV curve shown in Fig. 10 for the E–W configuration. The analysis demonstrates that the orientation and configuration of PV modules impact their performance throughout the day. Modules oriented towards the south capture more solar energy during peak sun hours, leading to increased module temperatures and higher power output. In contrast, East–West oriented modules are less effective around solar noon due to the sun's position, resulting in decreased solar irradiation capture, lower module temperatures, and reduced power output. It can also be seen from the table that the module temperature distribution for bifacial modules is better than their monofacial counterpart in all configuration with the exception for fixed south orientation. A possible explanation for this can be that generally bifacial modules have better solar irradiation absorption and conversion efficiency than monofacial modules resulting in less energy being dissipated as heat. However for the fixed south configuration, even if the bifacial module have better heat circulation characteristics, the inclined orientation could have negatively affected how air flows over and under the modules, influencing the cooling efficiency.

Finally, both module types, across all configurations, exhibit approximately the same performance in terms of fill factor (FF). The FF represents the ratio of the power measured at the maximum power point to the product of V_{oc} and I_{sc} at a particular irradiation level. For modules made from solar cells with identical intrinsic electrical properties (which is the case in this study), such as series resistance, shunt resistance, and diode ideality factors, the FF is expected to be similar between monofacial and bifacial modules. The increase in I_{sc} due to the bifacial gain in bifacial modules is accompanied by a corresponding increase in P_{mp} , which often scales proportionally, thereby keeping the FF relatively constant.

4.5. Bifacial gain

Bifacial gain is an important parameter for bPV systems, as it directly influences energy yield and system performance. This gain enhances the system's output, highlighting its significance in maximizing efficiency and return on investment in PV installations. The contribution of the rear side to overall energy capture is quantified through optical bifacial coefficient (BC_o). The results of this analysis for the four configurations throughout the year is shown in Fig. 11. As seen in this figure, the vertical bPV modules (MS01 and MS07) exhibit higher optical coefficient compared to the other configurations. This observation aligns with the expectation and is consistent with findings in the literature [20]. A possible explanation for this could be that vertical installations receive less direct irradiance on the front side of the module, while they benefit significantly from irradiance on the rear side (both reflected and diffuse components), which increases the optical gain. The most significant observation in this figure is the exceptionally high optical bifacial gain of the vertical E–W configuration. This result can be attributed to the prolonged exposure of this module

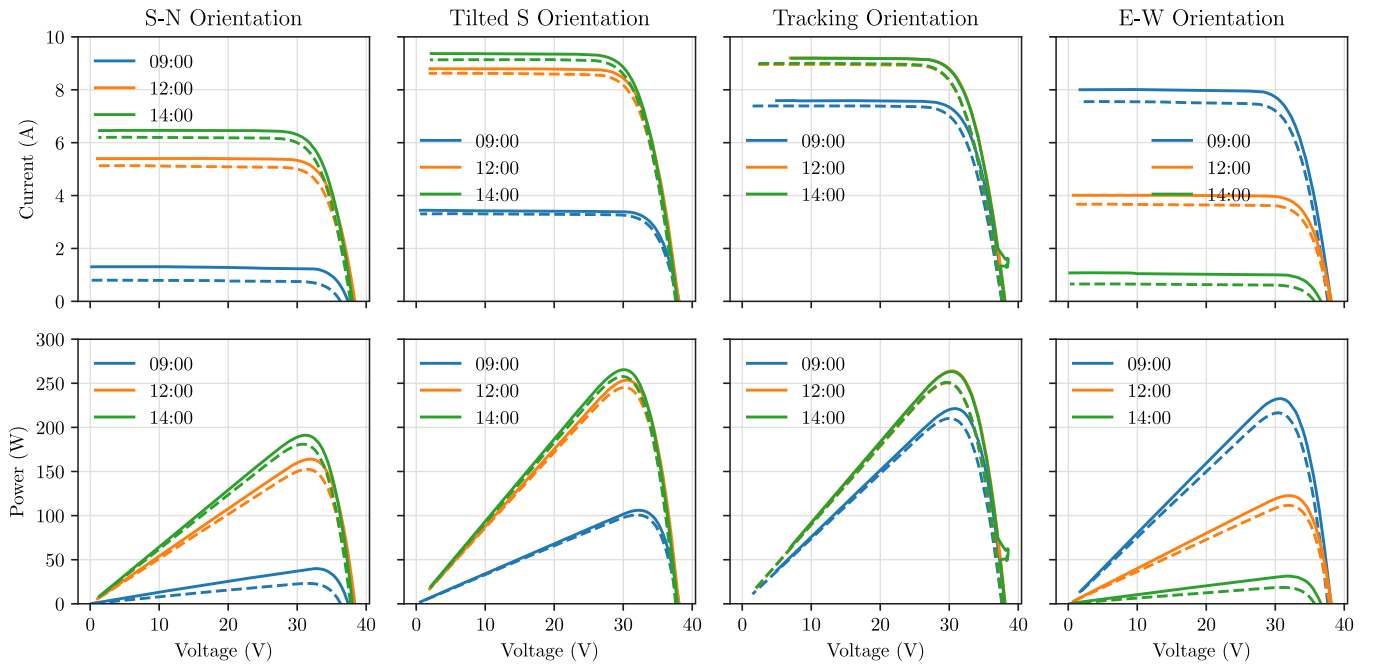


Fig. 10. IV and PV characteristic curves of the PV modules at Alpha Centauri on June 16, 2023, captured at different times of the day: 09:00, 12:00, and 14:00. Each plot represent the performance of the modules under various configurations. Dashed lines represent the monofacial modules.

Table 6

Outdoor characterization of PV modules at Alpha Centauri for a particular day. These data are obtained using automatic IV tracer.

Orientation	Time	Irradiance (W/m^2)		Temperature ($^{\circ}\text{C}$)		Vmp (V)		Imp (A)		FF (%)	
		Front	Rear	bPV	mPV	bPV	mPV	bPV	mPV	bPV	mPV
SN	09:00	98	192	22.9	24.2	32.88	31.48	1.22	0.74	82	80
	12:00	571	73	30.4	33.0	31.84	31.48	5.15	4.85	79	78
	14:00	681	66	34.1	38.5	31.07	30.77	6.15	5.88	78	77
Tilted S	09:00	373	49	29.4	29.0	32.20	32.01	3.29	3.15	81	81
	12:00	906	52	41.1	39.3	30.55	30.15	8.30	8.14	76	75
	14:00	960	49	41.5	43.5	30.08	30.01	8.83	8.59	75	75
Tracking	09:00	879	55	36.9	40.3	30.77	30.02	7.20	7.01	77	76
	12:00	973	65	26.6	40.3	30.35	29.58	8.70	8.47	75	74
	14:00	947	53	41.9	41.2	30.39	29.51	8.67	8.51	75	74
EW	09:00	783	54	37.3	40.9	30.71	30.52	7.58	7.10	76	76
	12:00	942	54	27.5	31.4	32.06	31.92	3.82	3.50	80	80
	14:00	945	48	25.7	26.2	31.81	30.92	0.99	0.60	80	79

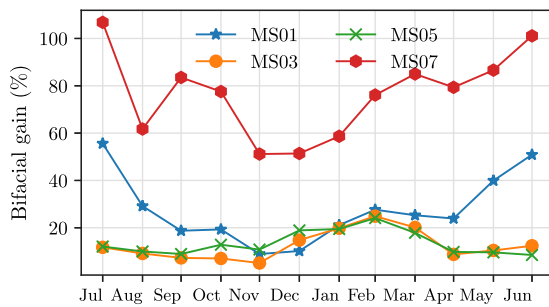


Fig. 11. Monthly averaged optical-level bifacial coefficients for bifacial modules with different configurations.

to direct sunlight throughout the day, especially during the summer season. The E-W and S-N bPV modules have a mean annual optical bifacial coefficient of 76% and 28%, respectively. On the other hand, tilted fixed south and tracking mode bPV modules have a relatively lower annual mean coefficient of 13% and 14%, respectively.

The relatively higher optical coefficients during the winter months (December, January, and February) for the tilted and tracking PV modules can be attributed to possible snow accumulation on the front side of the panel which resulted in partial shading and the higher albedo from the snowy surface that increased the rear irradiance. A typical example of the modules condition during winter is shown in Fig. 12. The annual distribution of the optical bifacial coefficient of the bPV modules is shown in the box plot shown in Fig. 13. It can be observed from this figure that the fixed tilted and tracking mode modules exhibit a relatively uniform distribution with low variation across all seasons of the year. An opposite pattern can be observed for the vertical configurations. A possible explanation for the observed patterns is the differing sensitivity of the module configurations to seasonal changes in solar position. Fixed tilted and tracking modules are designed to optimize the impact of seasonal variations, maintaining a more consistent performance year-round. Conversely, vertical configurations are more exposed to seasonal shifts in solar angle, resulting in greater fluctuations in their optical bifacial coefficient. This analysis shows that the contribution from the rear side of the module to the overall energy gain can be maximized using higher tilt angles. The red dot in Fig. 13 represents the optical-level bifacial coefficient values for June (50%) and July (56%) for the module oriented vertically in the



Fig. 12. Winter insight: PV module condition at Alpha centauri as of March 6th, 2023.

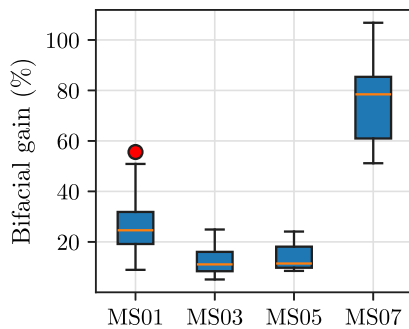


Fig. 13. Annual spread of the optical-level bifacial coefficient. The box represents the middle 50% of the data and the whiskers show the smallest and largest values. The orange line marks the median while the dot outside the whiskers indicate outliers.

south–north direction (MS01). These values differ significantly from the annual average value of 28% for this configuration, which is why they appear as red dots (outliers) in the plot. Since all the modules have the same specifications and the same bifaciality factor ($\phi = 0.65$), the module-level bifacial coefficient (BC_m) exhibits similar distribution and pattern as the optical bifacial coefficient.

The increased yield of bPV modules relative to mPV modules under identical operating conditions is expressed through the system bifacial gain (BGs). The outcomes of this analysis is shown in Fig. 14, encompassing all configurations for one year period. The average system bifacial gain ranges from 6% for tilted fixed systems (TS) to 47% for E–W configurations, showing considerable variability. This is in line with the optical coefficient discussed above. Notably, this gain is higher for vertically oriented modules. This suggests that the benefits of bifacial modules are maximized at steeper tilt angles, particularly nearing 90 degrees, in high latitude regions. Moreover, both tilted fixed and tracking modules exhibit relatively consistent performance with low standard deviation and spread compared to vertical modules, indicating consistent performance across different seasons. This observation is illustrated in Fig. 15. The red dot in Fig. 15 represents the system-level bifacial gain value for December (16%) for the module oriented tilted facing south (TS). This value is notably different from the annual average of 6% for this configuration, which is why it is shown as a red dot (outlier) in the plot. It should be noted that this is a limited study based on data from a one-year period; therefore, the results should be interpreted with caution, and further work using data collected over several years is needed.

5. Conclusion

In this paper, an empirical analysis of the performance of a bPV system in a high-latitude region was conducted using real-world data

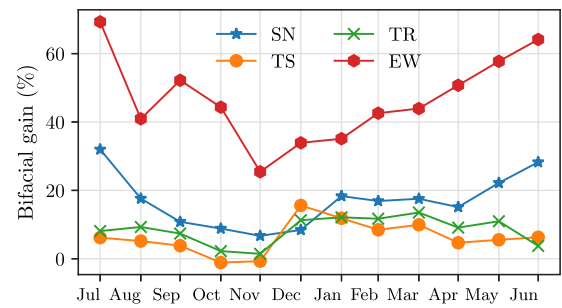


Fig. 14. Monthly averaged system-level bifacial coefficients/gains for different configurations.

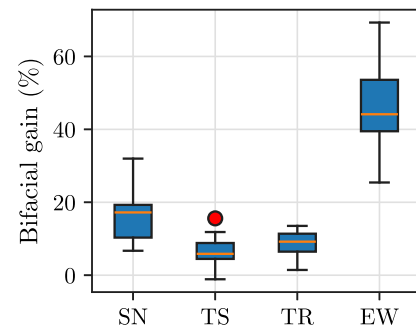


Fig. 15. Annual spread of the system-level bifacial gain. The box represents the middle 50% of the data and the whiskers show the smallest and largest values. The orange line marks the median while the dot outside the whiskers indicate outliers.

from an outdoor field laboratory in Trondheim, Norway. The findings from this study are summarized below.

- Multiple daily generation peaks due to varied orientations is a crucial characteristic, as it aligns more closely with the consumption profile. This factor plays a significant role in optimizing PV systems in BIPV scenarios, which is anticipated to experience rapid growth in high latitude regions.
- The correlation study indicates that bPV modules are particularly responsive to changes in irradiance. Both GHI and DNI similarly impact bifacial and monofacial modules. However, DHI has a more pronounced effect on vertically oriented bPV modules, making it a key factor to consider in high-latitude areas.
- The yield analysis reveals that springtime weather provides the most favorable conditions for PV performance in these regions. However, these findings are based on data from a one-year period and should be interpreted with caution. A bPV module with dual axis tracking yields only a 9% increase compared to a mPV module with the same tracking setup. Notable improvements in yield by choosing bPV over mPV are observed mainly in vertically oriented setups, particularly in east–west configurations.
- Bifacial modules typically demonstrate higher PR compared to monofacial modules across various configurations. Vertically oriented modules achieve high PR during the summer months, whereas fixed south-facing and tracking modules tend to excel in colder seasons.
- The IV–PV characterization reveals that while bifacial and monofacial PV modules with identical configurations exhibit similar characteristics, significant differences emerge across varied configurations, particularly in response to changes in solar irradiation and module temperature.

- The vertical bPV modules have higher optical bifacial coefficients compared to other configurations. On average, the E–W and S–N bPV modules achieve an annual optical bifacial gain of 76% and 28%, respectively. In contrast, the tilted fixed south and tracking mode bPV modules have a lower average annual optical coefficient of 13% and 14%, respectively. Nevertheless, these optical gains only result in a system level bifacial gain of 47%, 17%, 6.3%, and 8.4% for E–W, S–N, tilted fixed south, and tracking mode bPV modules, respectively.

Despite the estimated uncertainty (4.3%) in Section 3.4, the benefits of bifacial PV modules and vertical configurations can be highlighted in this work based on PR, annual yield, and bifacial gain. However, the implementation of these systems in high-latitude regions requires careful consideration of factors beyond these parameters. These include addressing challenges related to snow accumulation, snow and wind loads, and accounting for maintenance difficulties posed by harsh weather conditions. Additionally, accurate energy yield modeling that incorporates variable albedo, snow cover, and diffuse light conditions, as well as thorough economic viability assessments, is important. The empirical analysis and findings in this paper indicate the need to optimize bifacial PV systems further in high-latitude regions to facilitate their adoption. To strengthen and validate the findings of this paper, future research should consider the following:

- The amount of data used in the analysis may restrict the generalizability of some findings. Further studies that consider data collected over several years are recommended to capture long-term patterns.
- The lack of data related to snow cover during the winter season could limit the ability to explain certain results specific to the snowy winter months. Future work should incorporate this aspect.
- The uncertainty analysis in this paper focuses solely on instrument-related uncertainties and does not account for environmental factors, such as soiling from snow accumulation during winter. Further study should incorporate uncertainties related to environmental factors as well.

CRediT authorship contribution statement

Berhane Darsene Dimd: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Alfredo Sanchez Garcia:** Writing – review & editing, Writing – original draft, Validation, Software, Conceptualization. **Martin Bellmann:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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